Last Seen — HackHarvard 2026 Project Plan

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The pitch

Last Seen: the room that remembers, so you don't have to.

Last Seen is a camera-and-AI system for people with memory loss and the people who care for them, built from two iPhones and a laptop. It watches for the moment an object is picked up or set down, remembers where it went, and answers "Where's my inhaler?" out loud. At the same moment, a **memory frame** on the wall (an old phone mounted like a picture frame) lights up with a photo of the exact spot and a short clip of the person putting the item there.

The user never has to remember the system exists. It speaks up on its own when someone is searching, stops them at the door if they're leaving without their keys or inhaler, and lets a caregiver check from anywhere.

Track: Health. No custom hardware: everything runs on phones the team already owns and one laptop.

The one-sentence version for judges: *Babies learn object permanence at about eight months. People with dementia slowly lose it. We built a room that has it for them.*

The problem and who it's for

Losing things is a small annoyance for most people. For someone with memory loss it happens many times a day, and for some objects it's dangerous: a missing inhaler, EpiPen, or pill organizer is a health emergency, not an inconvenience.

Your teammate's figures (verify the latest numbers before the pitch): about 57 million people live with dementia worldwide, and more than 7 million Americans have Alzheimer's or a related condition.

Primary user: an older adult with early-stage dementia or mild cognitive impairment, living at home. They are still independent, which is exactly why they keep losing things. Someone in late-stage dementia is not the target.

Second user: the caregiver, usually an adult child or spouse. They field the "I can't find my glasses" calls, they worry about missed medication, and they are the person who would actually buy this.

Design rules that came out of our critique:

- **The user can't be expected to remember the system.** No app, no wake word they must recall, nothing to wear or charge.
- **The user may not talk to themselves.** Voice is one trigger among several, never the only one.
- **Never claim to see what the camera can't.** If an item goes under a blanket, say so honestly instead of pretending to know.
- **Life-critical items come first.** The inhaler matters more than the AirPods, and the pitch leads with it.

The five signature features

Each feature answers a specific weakness we found in earlier versions of the idea.

1. Touch-to-track memory

The system tracks anything a hand picks up, so it never has to guess what's worth remembering. When a hand closes on an object, that object is followed until it's set down, then its new spot is logged with a timestamp.

Every object gets one of five states, and the answers are always honest:

State	What happened	What Last Seen says
Placed	Set down somewhere visible	"Your keys are on the dresser. You put them there at 10:32."
Held	Currently in a hand	"You're holding them."
Hidden	Vanished while a hand was near a blanket, drawer, or bag	"Last seen going under the blanket on the bed."
Carried out	Left the camera's view in a hand	"You picked them up at 10:32 and walked out of the room."
Unknown	Not seen since enrollment	"I haven't seen them since this morning."

The **Hidden** state is the creative core of the project. We call it the **object permanence engine**: the system reasons that an object which disappears beside a blanket is probably under it, the same way a person would.

2. Memory frame: the answer comes to them

A second iPhone (or an old iPad) sits on the wall by the bed or the door, like a picture frame. Most of the time it shows a calm clock. When anyone asks where something is, it switches to full screen: a photo of the room with the item circled, the item's name in large text, and the clip of it being put down, playing on loop.

The person never picks up a device or opens an app. The answer appears where they already are. In the caregiver app, the item also gets a highlight that follows it on the live video as it moves.

3. Search sense: it notices before you ask

The camera also tracks the person's pose. Searching has a recognizable pattern: standing, moving between areas without settling, returning to the same spot, lifting cushions or opening drawers. When that pattern lasts long enough, Last Seen speaks up gently and **makes a guess** instead of reading out a list:

- Standing near the door → "Looking for your keys? They're on the dresser."
- Near medication time → "Your inhaler is on the nightstand."
- Otherwise → the item they handled most recently.

Voice works too: "Where are my glasses?", "Have you seen the EpiPen?", spoken by anyone in the room. No wake word needed, since the system listens for those phrases specifically.

4. Rescue mode for critical items

Inhalers, EpiPens, and medication are tagged as critical. They get three extra protections:

- **Instant answer:** "Where's the EpiPen?" skips any small talk, answers immediately, and puts the spot on the memory frame full screen. In an emergency it's often a family member asking, not the patient.
- **Door check:** walking toward the door while a critical item stays behind triggers "Wait, your inhaler is still on the nightstand."
- **Caregiver alerts:** if a critical item hasn't been seen for a set time, or the pill organizer hasn't been opened by evening, the caregiver gets a real text with the photo, through Photon.

5. Proof, not just an answer

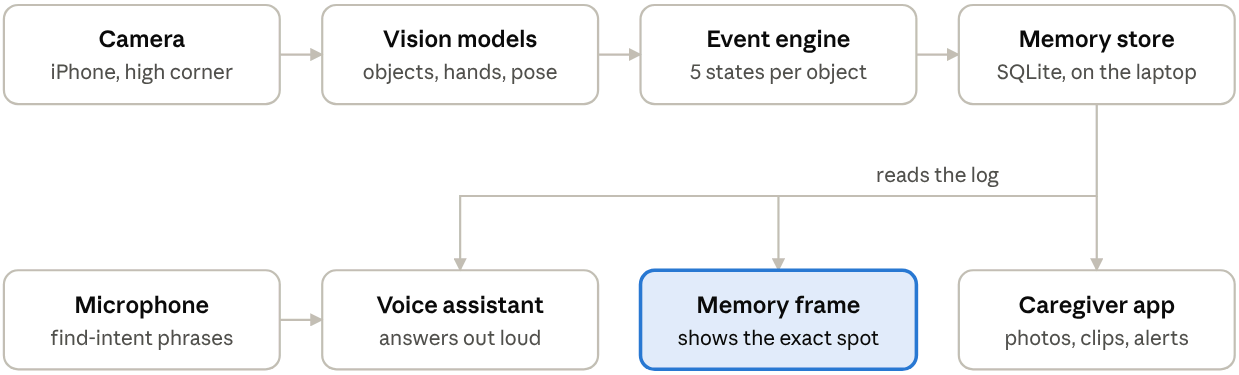
Every answer comes with a photo of the room with the item's spot highlighted, which your teammate suggested. Add one more thing: a **3-second clip of the moment the person put it down.** Memory research consistently finds recognition is easier than recall, so seeing yourself place the keys helps the memory click back into place. It also builds trust that the system is right.

The caregiver app shows the same photos, a timeline of events, and the status of critical items.

How it works

Everything runs on one laptop in the room. The camera sends frames in, an event engine turns them into a small database of object events, and three outputs read from that database: the voice assistant, the memory frame, and the caregiver app.

One local event log feeds every answer



Last Seen architecture · 8 parts, all on one laptop

The vision pipeline writes only state changes to the memory store; the voice assistant, the memory frame, and the caregiver app all answer from that one log.

Layer	Tool	Notes
Camera	iPhone 1, streamed to the laptop with Continuity Camera (Mac) or DroidCam	Mounted high in a corner. 5 to 10 frames per second is plenty.
Object detection	YOLO-World (open-vocabulary)	Finds "keys", "inhaler", "glasses" from text prompts. Standard YOLO can't: its COCO classes have no keys, glasses, or wallets.
Fallback detector	YOLOv8/YOLO11 fine-tuned on your demo objects	Label about 150 photos per object in Roboflow. Do this only if YOLO-World misses small items.
Item identity	DINOv2 or CLIP embeddings	Enroll each item once ("this is *my* inhaler") by holding it up, so it's told apart from similar objects.
Hands	MediaPipe Hands	A hand box overlapping an object box for a few frames = picked up.
Person pose	MediaPipe Pose	Feeds the search-sense score: standing, moving, revisiting zones.

Layer	Tool	Notes
Event engine	Python state machine, one per object	Placed, Held, Hidden, Carried out, Unknown. Writes events only when state changes.
Zones	Polygons drawn once on a snapshot	"Nightstand", "bed", "dresser", "door". A point-in-polygon check names each location.
Memory store	SQLite	Tables: items, events, clips. Small and local.
Speech in	faster-whisper (local)	Listens for find-intent phrases, discards everything else.
Reasoning	LLM routed through Chalk's AI gateway	Turns a question into a database lookup and a natural sentence, using events and zones as context. Sends text only.
Speech out	macOS `say` or Piper TTS	Calm, slow voice.
Memory frame	iPhone 2 or an old iPad, running a full-screen web page	Gets each answer from the laptop over a WebSocket and shows it full screen.
Caregiver app	React or Next.js, WebSocket to a FastAPI backend	Live room photo with highlights, timeline, critical-item status. Alerts and text-back questions go through a Photon 2.0 number (iMessage/SMS), plus a voice call for the door check.

Camera placement. Prop iPhone 1 high on a shelf or bookcase, angled down at the mock room, and keep it plugged in. Keep the room small (a nightstand, a bed or couch, a tray, a "door") so small items take up enough of the frame to detect. Turn off auto-lock so the stream doesn't stop.

Privacy. Video never leaves the laptop and is never saved, apart from short event clips of hands and objects. Only text goes to Chalk and Photon, plus the one photo and clip sent to the caregiver. Those are encrypted and deleted after 24 hours. Bathrooms and

changing areas are excluded with privacy zones, and a privacy-mode button pauses everything. Say all of this in the pitch: a camera in a bedroom is the first thing judges will worry about.

Equipment

There are no parts to buy or wire. Everything below is something the team already owns or can borrow.

Item	Role
iPhone 1	The room camera, propped high on a shelf and plugged in
iPhone 2 (or an old iPad)	The memory frame, leaned on a shelf or taped to the wall
A third phone	Runs the caregiver app, to receive the alert in the demo
Laptop (a Mac with Apple silicon is ideal)	Runs every model and the database
Chargers and cables for each phone	Keeps the camera and frame alive through the demo
Mock room props	A small table or nightstand, a blanket, a tray, a taped "door" on the floor
Demo objects	Keys on a large keychain, a glasses case, an empty inhaler, a pill organizer, a trainer EpiPen

Use an **empty or trainer inhaler and EpiPen** for the demo. Never bring real medication.

The real product would use small cameras that blend into each room and a dedicated display. Say this in one sentence; don't build it.

The 3-minute demo

The demo is the product. Rehearse it at least ten times, including what you'll do if a step fails.

1. **The hook (0:00–0:20).** "Babies learn object permanence at eight months. For 57 million people with dementia, it slowly goes away. We built a room that remembers for them."
2. **Enrollment (0:20–0:40).** Ask a judge for something of theirs, like sunglasses. Hold it up to the camera: "These are your sunglasses." It's enrolled in seconds. This proves nothing is hardcoded.
3. **Living in the room (0:40–1:10).** A teammate plays the user. They put the judge's sunglasses on the nightstand, set their keys on the tray, and slide the inhaler under the blanket. The caregiver app on the screen updates live, with each item's highlight following it as it moves.
4. **The ask (1:10–1:40).** The judge says out loud, "Where are my sunglasses?" The room answers, and **the memory frame on the wall lights up with the nightstand circled and the clip of the sunglasses being put down.** Pause and let that land; this is the moment judges will remember.
5. **The honest answer (1:40–2:00).** Ask, "Where's the inhaler?" The system says it was last seen going under the blanket on the bed, and the frame circles the bed. Lift the blanket. This shows the object permanence engine.
6. **Rescue mode (2:00–2:30).** The teammate walks toward the taped door with the inhaler still in the room. The room stops them: "Wait, your inhaler is still on the bed." A judge's phone, standing in for the caregiver, rings with a real call from Last Seen's number, then gets a text with the photo. The judge texts back "where's her inhaler?" and gets the answer.
7. **Close (2:30–3:00).** Explain privacy in one sentence (local only, no video saved). End with: "She doesn't have to remember. The room does."

Backups if something breaks: keep a pre-recorded video of the full demo on the laptop. If the memory frame loses its connection, show the same answer on the laptop screen. If voice fails, type the question into the app.

Team split and 36-hour plan

This assumes a team of four. With three, merge the voice and app roles.

Person	Owns	Done means
A: Vision	Detection, enrollment, hand tracking, the event state machine	Moving an object updates its location in the database within 2 seconds

Person	Owns	Done means
B: Brain and voice	Speech in and out, question answering through Chalk, the Chalk eval set, search sense, rescue-mode rules	"Where's my inhaler?" gets a correct spoken answer
C: Frame and room	Memory frame page, camera stream and placement, event clips, highlighted photos, mock room	The frame shows the circled spot and the clip within 2 seconds of a question
D: App and pitch	Caregiver app with live highlights, Photon texts and calls, slides, demo script	The app shows the live room and alerts; the pitch is rehearsed

Checkpoints:

1. **Hour 4:** iPhone camera streaming to the laptop, detector seeing all demo objects, frame page showing a test answer.
2. **Hour 12:** pick-up and put-down events land in the database. *If detection still fails on small items, switch to the fine-tuned fallback now.*
3. **Hour 20:** voice question → spoken answer → memory frame shows the spot. **This is the full demo loop; everything after is polish.**
4. **Hour 26:** Hidden and Carried-out states, door check, Photon text alerts and text-back.
5. **Hour 30:** search sense, event clips, Photon voice call, Chalk eval run. Feature freeze.
6. **Hours 30–36:** record the backup video, rehearse the demo ten times, write the submission.

Cut list, in order, if you fall behind:

1. Search sense (keep voice questions and the door check)
2. Event clips (keep the highlighted photos)
3. Local LLM (use rule-based answers)
4. Photon voice call (keep the text alerts, which the Photon prize needs)

Never cut: the memory frame, the honest "under the blanket" answer, or the door check. Those three are what make this project different from every other object finder.

Standing out, judge questions, and risks

Why this beats what already exists

Compared with	What it does	What Last Seen adds
AirTags and Tile	Find tagged items with a phone	No tags, no phone, nothing to charge; works on any object
ReCall (Hack the North project)	Sends webcam snapshots to a cloud vision model, answers from text descriptions	Tracks individual objects locally, detects events, shows the exact spot, speaks up on its own, built for dementia
Smart speakers	Answer questions you remember to ask	Notice searching and leaving the house without being asked

If you fall back to "send frames to a vision model and ask it," you've rebuilt ReCall. The local event engine is the difference, so make sure it works.

Questions judges will ask

- **"Isn't a camera in someone's bedroom creepy?"** Everything is processed locally, no video is saved, short clips auto-delete in 24 hours, there's a physical off switch, and privacy zones exclude bathrooms. The caregiver sees object locations, not a live feed.
- **"What if it's wrong?"** It says how confident it is and shows the photo, so you can check. It never claims to see through a blanket.
- **"What about objects carried to another room?"** It says "you carried them out at 10:32." The real product has one camera per room, so the trail continues.
- **"Who pays for this?"** Families buying for a parent, and later senior living facilities and in-home care agencies.
- **"Did you talk to real users?"** Have an answer. Even one conversation with a caregiver before the event counts.

Risks and fallbacks

Risk	Fallback
Detector misses small items at room distance	Prop the camera closer and higher, use larger demo objects, fine-tune on your objects
Phone camera stream lags or drops	Connect the camera phone by USB cable instead of Wi-Fi; keep it charging
Venue Wi-Fi drops	Put the laptop and frame phone on a phone hotspot
Noisy room breaks voice	Use a wired or AirPods mic close to the speaker; typed questions as a backup
Team runs out of time	Follow the cut list; the Hour 20 loop alone is a strong demo

Sponsor prizes

Go for two: **Photon 2.0** and **Chalk**. Both use pieces Last Seen already needs. Every other challenge would mean bolting something on.

Photon 2.0: main target

Prizes: Winner ($400 + $300 credits + interview fast-track), and the Best Voice Agent sub-prize ($300 + $100 credits). The project must live on at least one Photon channel.

- **Alerts as real texts.** Last Seen gets its own phone number. Rescue-mode alerts arrive by iMessage or SMS: "Mom's inhaler hasn't been seen since 2pm," with the highlighted photo attached.
- **The caregiver can text back.** "Where's her inhaler?" gets the answer, the photo, and the clip. The caregiver app becomes a conversation.
- **Voice for the sub-prize.** When the door check fires, Last Seen places a real call to the caregiver. Any family member can also call the number in an emergency and ask out loud, which fits the point that it's often a relative asking about the EpiPen.
- **Owner:** Person D. Attend the Friday 4–5 PM workshop.

Chalk: second target

Prize: $1,500+ in Apple gift cards. Judged on impact, use of data, and how it's evaluated.

- **The data:** the object event log and the room zones are the real-world context. Route the question-answering model through Chalk's AI gateway with that data as its input.
- **The evaluation is where you stand out.** Record a dozen scripted scenarios and score Last Seen on each: did it give the right location and the right state? Report a single headline number, such as "correct on 11 of 12, and an honest 'I don't know' on the 12th" (that's an example; report whatever you actually get).
- **Scenarios to include:** keys moved to the tray (Placed), inhaler slid under the blanket (Hidden), keys carried out of the room (Carried out), an item held while asked (Held), an item never moved (Placed), and two similar items side by side (identity).
- **Owner:** Person B.

Skipping

- **Daytona and Factory Droid:** both must be a meaningful part of the running product, and nothing here needs agent-written or untrusted code run in the cloud.
- **SpaceXAI and Capital One Nessie:** require space data or banking.
- **ASUS Content Challenge:** tied to the Sustainability track. Ask an organizer whether Health teams can enter; if so, a short vlog of building the mock room is cheap content.

Keep the privacy story true: video still never leaves the laptop. Only text (events, questions, answers) goes through Chalk and Photon, plus the single highlighted photo and clip sent to the caregiver. Say this in one sentence in the pitch.

Before the hackathon

Check HackHarvard's rules first: most hackathons require code to be written during the event, but research, testing, and buying parts beforehand are normally fine.

- [ ] Test YOLO-World on an iPhone stream with keys, glasses, and an inhaler from 2–3 meters. **This decides whether the project is viable.** If it fails badly, plan for fine-tuning.
- [ ] Test streaming the iPhone camera to the laptop (Continuity Camera or DroidCam) and check the delay.
- [ ] Attend the Photon 2.0 workshop (Friday, 4–5 PM) and get Last Seen's phone number working on iMessage or SMS.
- [ ] Set up Chalk access and confirm how its AI gateway and eval tools work.

- [ ] Find an old phone or iPad for the memory frame, plus chargers and cables for every phone.
- [ ] Ask an organizer whether Health-track teams can enter the ASUS content challenge.
- [ ] Talk to at least one caregiver of someone with memory loss for ten minutes. Ask how they search for things and which items matter most.
- [ ] Verify the dementia statistics against a current source for the slides.
- [ ] Buy or borrow a trainer EpiPen and an empty inhaler.
- [ ] Agree on roles and the Hour 20 demo loop with the team.